

STAT 341 - Assignment 4 - Spring 2025

Due: Friday, Aug 1, 2025 at 11:59

Your assignment must be submitted by the due date listed at the top of this document, and it must be submitted electronically in .pdf format via Crowdmark. This means that your responses for different questions should begin on separate pages of your .pdf file. Note that your .pdf solution file must have been generated by R Markdown. Additionally:

- For mathematical questions: your solutions must be produced by LaTeX (from within R Markdown). Neither screenshots nor scanned/photographed handwritten solutions will be accepted – these will receive zero points.
- For computational questions: R code should always be included in your solution (via code chunks in R Markdown). If code is required and you provide none, you will receive zero points.
- For interpretation questions: plain text (within R Markdown) is required. Text responses embedded as comments within code chunks will not be accepted.

Organization and comprehensibility is part of a full solution. Consequently, points will be deducted for solutions that are not organized and incomprehensible. Furthermore, if you submit your assignment to Crowdmark, but you do so incorrectly in any way (e.g., you upload your Question 2 solution in the Question 1 box), you will receive a 5% deduction (i.e., 5% of the assignment's point total will be deducted from your point total).

QUESTION 1 [20 marks] - Multiple Testing and Transformed Data

In class, we did random mixing of two groups to get a distribution of the attributes of the sampling distribution if the null hypothesis “these two subpopulations are indistinguishable”. This was part of a set of randomization tests to estimate p-values for different attributes about the subpopulations, and we used a “minimum of p-values” method to account for the multiple testing problem when those attributes might be correlated.

In this assignment we will apply a similar random mixing approach, and the same ‘minimum of p-values’ approach to testing for a correlation between two variables under different transformations. To do this, we want some version of random mixing that would produce a distribution of attributes under the null hypothesis “variables x and y are independent”. We will do that by starting from an observed set of ordered pairs $(x_i, y_i), i = 1, \dots, 24$ and producing random pairings by randomly mixing the y values with the x values.

The following function will be valuable to you in this question.

```
randomPairing = function(x,y)
{
  xrand = sample(x)
  yrand = sample(y)

  return(list(x=xrand, y=yrand))
}
```

The following dataset will also be useful.

```
df = read.csv("Stat341_S25_LinkedIn_Dataset.csv")
head(df)
```

##	AmpmEst	Comments	Reacts	Impressions	Members	Hashtags	Prompt	Picture	DOW
## 1	pm	0	16	876	410	Intermixed	No	No	Fri
## 2	pm	2	7	835	396	End	Yes	No	Mon
## 3	am	1	7	801	405	End	No	No	Wed
## 4	am	0	4	583	318	End	No	No	Wed
## 5	pm	11	16	4718	2829	End	Yes	Yes	Fri
## 6	pm	0	11	765	376	End	Yes	No	Mon

```
dim(df)
```

```
## [1] 24 9
```

- a) (2 marks) Use the `cor` function to find the Pearson correlation between **Impressions** (y), and **Reacts** (x). Put this estimate into the t-test for testing if a Pearson correlation is zero. Use a two-tailed test, and compare against the T_{n-2} distribution.

$$H_o : \rho = 0, \quad H_A : \rho \neq 0 \quad t_{n-2,obs} = r \sqrt{\frac{n-2}{1-r^2}}$$

- b) (3 marks) Apply the transformations in the following table to both variables of the data and compute the sample correlation r as well as the t-score and p-value as you did in part a.

Transformation of x,y	Correlation r	Test Statistic t	p-value
x, y			
log(x+1), log(y+1)			
sqrt(x), sqrt(y)			
-1/(x+1), -1/(y+1)			

c) (4 marks) Use the `randomPairing` function to produce a random pairing of the untransformed x and y. Plot in a 2x2 grid using `par(mfrow = c(2,2))` the following plots:

- Top left - Histogram of observed x, `df$Reacts`
- Top right - Histogram of observed y, `df$Impressions`
- Bottom left - Histogram of randomPairing x, `'rp$x'`
- Bottom right - Histogram of observed x, `'rp$y'`

Very briefly comment on the suitability of a transformation in each situation.

- d) (4 marks) Generate 5000 random Pairings of the untransformed x and y, and find the t-score of each one. Plot a histogram of the 5000 t-scores with a red (`col="Red"`) vertical line of width (`lwd = 3`) with `abline(v=..., col="Red")`.
- e) (1 mark) Empirically estimate the p-value from the observed t-score, and the 5000 random pairings. (Remember that you're doing a two-tailed test)
- f) (4 marks) For the four transforms in part b (including the identity transformation), on 5000 random pairings, find the theoretical p-values against the null of no correlation for each transformation. Draw a scatterplot of the p-values of the untransformed random pairings in the x, and the log-transformed random pairings in the y *for the first 500 values of each*. (Note: This means that for each random pairing you should find the correlation four times and get four p-values. There should be 5000 pairings in total, not 20,000.)
- g) (2 marks) For each of the 5000 sets of p-values in part f, find the minimum of the four p-values from that set. Report the proportion of times that the minimum p-value from the random pairings was lower than the minimum p-value from the observed data (you should be able to get the observed minimum p-value from your answer in part b)

This proportion is your estimate of the p-value adjusted for testing across multiple transformations.

You may find either the `pmin()` function or the `apply()` function valuable.

QUESTION 2 [15 marks]: A copy, of a copy, of a copy, of a legal document.

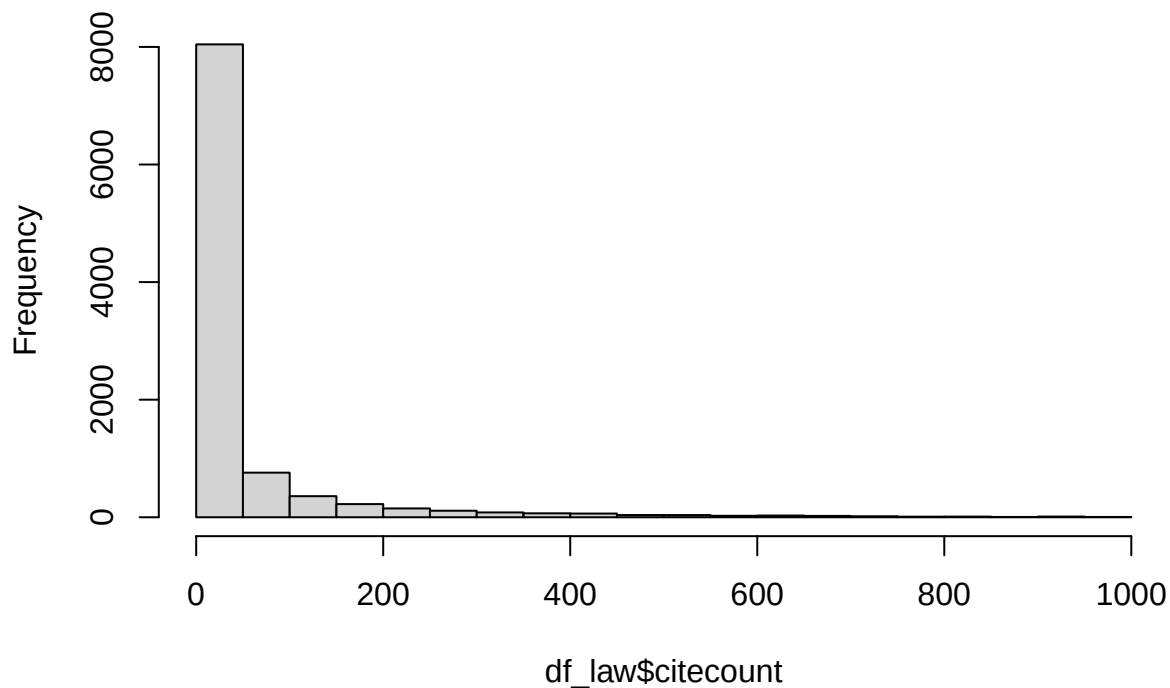
In this question you will explore how well (or poorly) a set of bootstrap resamples and double bootstrap re-resamples preserves the sampling distribution of the original study population.

The y values of interest are the number of citations found in-network (that is, a small portion of the Canadian Legal Landscape) from a 2015 snapshot of the Supreme Court of Canada decisions found from CanLII (Canadian Legal Information Institute).

The dataset where you can find this is `Lawnodes.csv` and the variables of interest are the (in-network) citation count `Citecount`, and the identifying code, `ID`.

```
df_law = read.csv("Lawnodes.csv")
#df_law$id = 1:nrow(df_law)
hist(df_law$citecount)
```

Histogram of `df_law$citecount`



We are going to use the following samples of size 5, 10, 20, and 50.

```
set.seed(341)
df_S05 = df_law[sample(1:nrow(df_law), 5) , c("ID", "citecount")]
df_S10 = df_law[sample(1:nrow(df_law), 10) , c("ID", "citecount")]
df_S20 = df_law[sample(1:nrow(df_law), 20) , c("ID", "citecount")]
df_S50 = df_law[sample(1:nrow(df_law), 50) , c("ID", "citecount")]
```

- a) (2 marks) Find the sample mean, variance, and number of unique values (`unique-Ncite`) of `citecount` in the population and each of the samples, and fill the following table. You may use `mean()`, `var()`, and `length(unique())` to find these.

Dataset	mean	variance	unique-Ncite
Population			
Sample size 5			
Sample size 10			
Sample size 20			
Sample size 50			

- b) (2 marks) From the sample of size 5, resample $B = 1000$ times with replacement using the following code. Report in a 2x2 histogram, the following: resampled means, variances, number of unique citation counts, number of unique IDs.

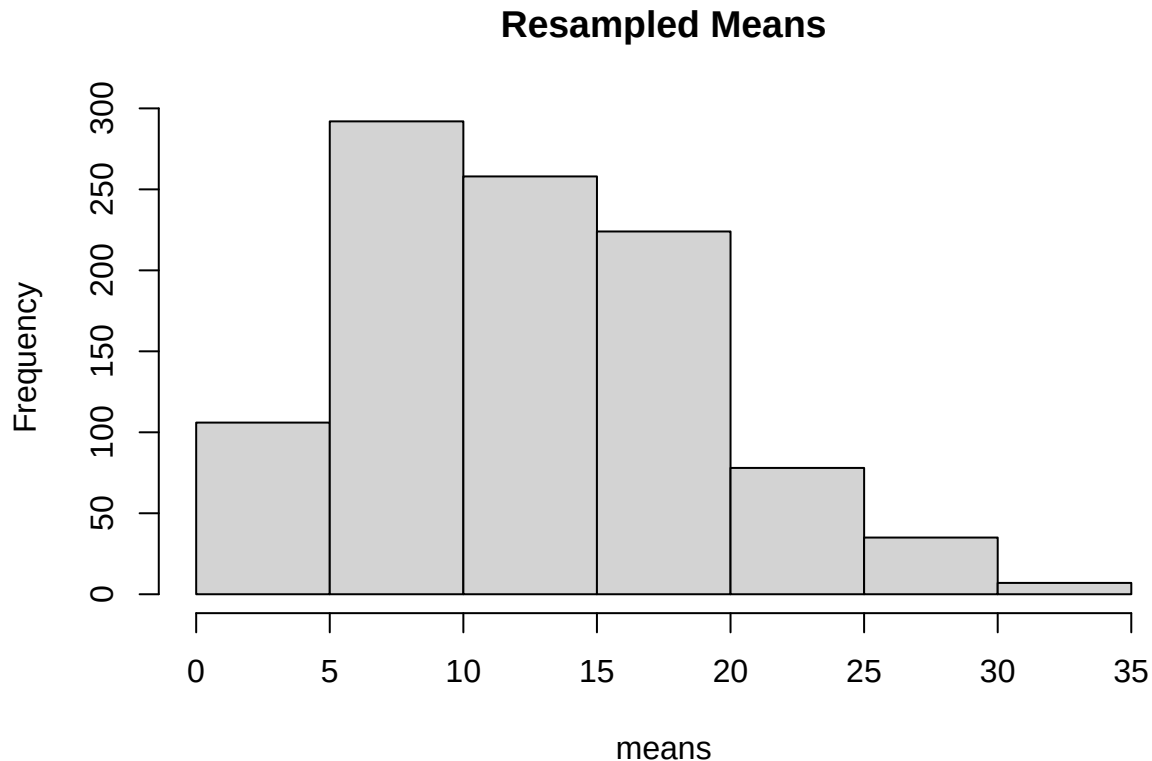
```

B=1000
S_idx = 1:5
n = length(S_idx)
means = rep(NA,B)
vars = rep(NA,B)
uniques_Ncite = rep(NA,B)
uniques_ID = rep(NA,B)

for(k in 1:B)
{
  Sstar_idx = sample(S_idx, n, replace = TRUE)
  means[k] = mean(df_law$citecount[Sstar_idx])
  #var[k] = ...
  #uniques_Ncite[k] = ...
  #uniques_ID[k] = ...
}

#par(mfrow = c(2,2))
hist(means, main="Resampled Means")

```



```
#hist(vars, main="Resampled Variances")
#hist(uniques_Ncite, main="Resampled Unique Citation Counts")
#hist(uniques_ID, main="Resampled Unique Legal Decision")
par(mfrow = c(1,1))
```

c) (2 marks) Repeat part b for the sample of size 50.

d) (3 marks) From the sample of size 5, resample $B = 1000$ times with replacement, and re-resample each of these $D = 100$ times using the following code. Report in a 2x2 histogram, the following: re-resampled means, variances, number of unique citation counts, number of unique IDs.

```
B=1000
D=100
S_idx = 1:5
n = length(S_idx)
means = rep(NA,B*D)
vars = rep(NA,B*D)
uniques_Ncite = rep(NA,B*D)
uniques_ID = rep(NA,B*D)

for(k in 1:B)
{
  Sstar_idx = sample(S_idx, n, replace = TRUE)

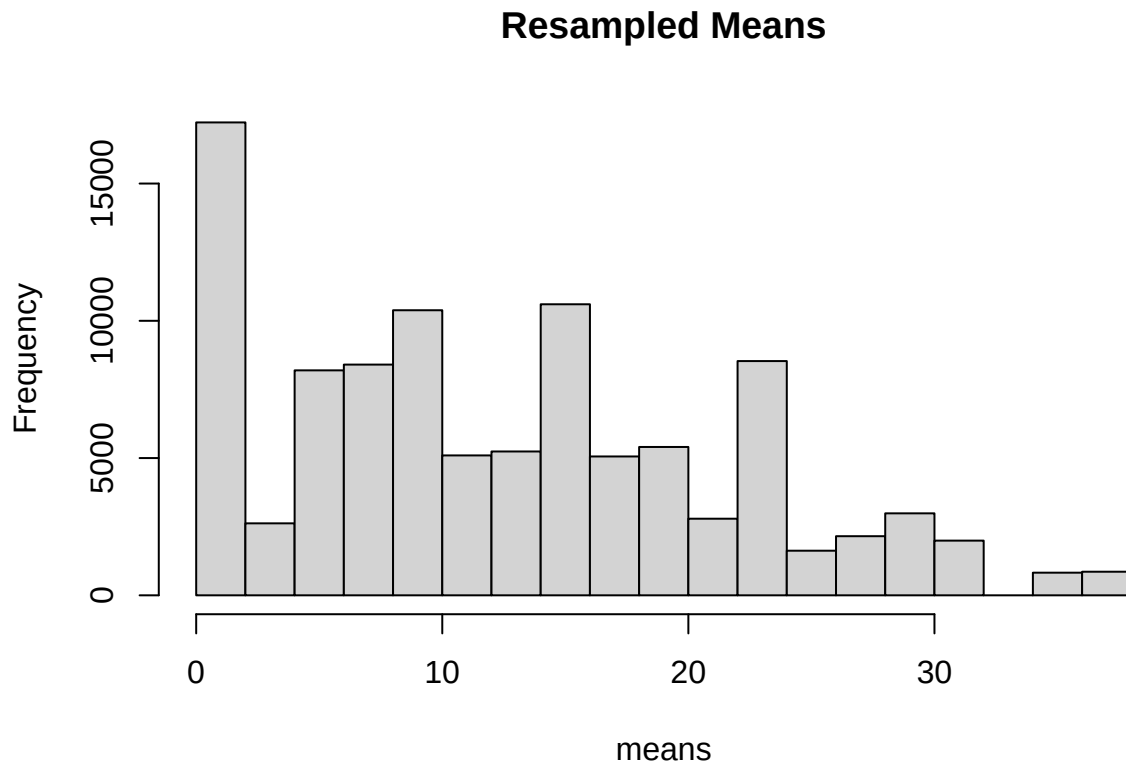
  for(l in 1:D)
  {
    Sstarstar_idx = sample(Sstar_idx, n, replace=TRUE)
    means[l + (k-1)*D] = mean(df_law$citecount[Sstarstar_idx])
    #var[l + (k-1)*D] = ...
```

```

#uniques_Ncite[l + (k-1)*D] = ...
#uniques_ID[l + (k-1)*D] = ...
}
}

#par(mfrow = c(2,2))
hist(means, main="Resampled Means")

```



```

#hist(vars, main="Resampled Variances")
#hist(uniques_Ncite, main="Resampled Unique Citation Counts")
#hist(uniques_ID, main="Resampled Unique Legal Decision")
par(mfrow = c(1,1))

```

- e) (2 marks) Repeat part b for the sample of size 50.
- f) (2 marks) Comment on how well (or poorly) the sample of size 5 maintains its ability to describe the data when being resampled and re-resampled?
- g) (2 marks) Comment on how well (or poorly) the sample of size 50 maintains its ability to describe the data when being resampled and re-resampled? Include a comparison to the size 5 sample, if appropriate.

QUESTION 3: Bootstrap Confidence Intervals [25 marks]

In this question, you will turn your focus to (point and interval) estimation of the *treatment effect*, a quantification of the impact on **dissent** on **citecount**. That is, the effect that the existence of a minority dissenting opinion in the Supreme Court of Canada over a decision has on the number of times the decision

is cited later (within the network measured). Treatment impact can be quantified in a variety of ways; here you will focus on the *average treatment effect*

$$\text{ATE} = \bar{y}_T - \bar{y}_C$$

and also the percent treatment effect, referred to in A/B testing settings as *lift*:

$$\text{lift} = \frac{\bar{y}_T - \bar{y}_C}{\bar{y}_C}$$

where

$$\bar{y}_T = \frac{1}{n_T} \sum_{u \in \mathcal{S}_T} y_u \quad \text{and} \quad \bar{y}_C = \frac{1}{n_C} \sum_{u \in \mathcal{S}_C} y_u$$

are the average citation counts in the treatment (dissenting) and control (unanimous) groups, respectively.

(a) (4 marks) Consider the following simple linear regression model

$$y_u = \alpha + \beta x_u + r_u, \quad u \in \mathcal{S} = \mathcal{S}_T \cup \mathcal{S}_C$$

where y_u represents the `citecount` of decision u and x_u is a treatment assignment indicator defined as follows:

$$x_u = \begin{cases} 1 & \text{if decision } u \text{ had dissent} \\ 0 & \text{if decision } u \text{ was unanimous} \end{cases}$$

Earlier in the course, it was established that the least squares estimates of α and β are

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} \quad \text{and} \quad \hat{\beta} = \frac{\sum_{u \in \mathcal{S}} (x_u - \bar{x})(y_u - \bar{y})}{\sum_{u \in \mathcal{S}} (x_u - \bar{x})^2}$$

where $\bar{x} = \sum_{u \in \mathcal{S}} x_u / n$, $\bar{y} = \sum_{u \in \mathcal{S}} y_u / n$, and $n = n_T + n_C$.

For x_u and y_u as defined above, show that

$$\hat{\alpha} = \bar{y}_C \quad \text{and} \quad \hat{\beta} = \bar{y}_T - \bar{y}_C$$

and hence that $\text{ATE} = \hat{\beta}$ and $\text{lift} = \hat{\beta} / \hat{\alpha}$.

****Hints:****

```
* $\sum_{u \in \mathcal{S}} x_u = n_T$
* $\sum_{u \in \mathcal{S}} y_u x_u = \sum_{u \in \mathcal{S}_T} y_u = n_T \overline{y}_T$
* $x_u^2 = x_u \sim \text{for all } u$
```

(b) (2 marks) Using the `lm()` function, fit the linear regression model above and calculate the ATE and lift for this (post-hoc) A/B test.

- (c) (2 marks) By resampling $\mathcal{S}$ with replacement, construct $B = 1000$ bootstrap samples $\mathcal{S}_1^*, \mathcal{S}_2^*, \dots, \mathcal{S}_{1000}^*$. Print out your code, not the 1000 samples. Note that the units in $\mathcal{S}$ should be regarded as the (x_u, y_u) pairs. Thus the pairs themselves should be resampled with replacement.
- (d) (6 marks) For each of the $B = 1000$ bootstrap samples in part (c), use `lm()` to calculate the least squares estimates $\hat{\alpha}^*$ and $\hat{\beta}^*$ and hence ATE^* and hence lift^* in each bootstrap sample $\mathcal{S}_b^*$, $b = 1, 2, \dots, B$. Construct two histograms, one of the ATE^* values and the other of the lift^* values. Include a vertical line representing ATE from (b) on the first histogram and a line representing lift from (b) in the second. Be sure to informatively label your plots.
- (e) (2 marks) Calculate 95% confidence intervals for ATE and lift using the naive normal theory approach.
- (f) (2 marks) Calculate 95% confidence intervals for ATE and lift using the percentile method.
- (g) (2 marks) Calculate 95% confidence intervals for ATE and lift using the bootstrap- t approach. You should use the `bootstrap_t_interval_new()` function included in the Appendix. Note that this is a modified version of the `bootstrap_t_interval()` function from class that accommodates a data frame (instead of a vector) for the input `S`. For the input `a`, you will also need to write functions that calculate ATE and lift given a data frame that includes columns for `citecount` and `dissent`. Please use $B = 100$ and $D = 100$.
- (h) (3 marks) This question concerns advantages and disadvantages associated with the various methods of confidence interval calculation you've explored.
- i. (1 mark) List one advantage and one disadvantage of naive normal theory intervals.
 - ii. (1 mark) List one advantage and one disadvantage of percentile method intervals.
 - iii. (1 mark) List one advantage and one disadvantage of bootstrap- t intervals.

OPTIONAL QUESTION 1: Practice Interview Questions [0 points]

Here are four very common data science interview questions related to the material covered on this assignment. In preparation for your own interviews, or the final exam perhaps, think about how you would answer them. In all cases, imagine you're discussing these things with the Senior Data Scientist who is interviewing you. You do not need to submit answers for these.

- (a) [0 points] In your own words, explain what a p-value is.
- (b) [0 points] In your own words, explain what the randomization test is, and why it's useful.
- (c) [0 points] In your own words, explain what it means to be “95% confident” in a 95% confidence interval.
- (d) [0 points] In your own words, explain what resampling methods like the bootstrap are, and why they are useful.

USEFUL FUNCTIONS

Note that in the functions below the input `pop` must be a list containing two elements, `pop1` and `pop2` which are themselves data frames containing the data corresponding to sub-populations $\mathcal{P}_1$ and $\mathcal{P}_2$, respectively. Also note that `calculatePVMulti` below works exactly the same as `calculateSLmulti` from Section 4.2.3 of the lecture notes.

```
calculatePVMulti <- function(pop, discrepancies, M_outer, M_inner) {  
  
  ## Local function to calculate the significance levels over the discrepancies  
  ## and return their minimum  
  
  getPVmin <- function(basePop, discrepancies, M) {  
    observedVals <- sapply(discrepancies, FUN = function(discrepancy) {  
      discrepancy(basePop)  
    })  
  
    K <- length(discrepancies)  
  
    total <- Reduce(function(counts, i) {  
      # mixRandomly mixes the two populations randomly, so the new sub-populations  
      # are indistinguishable  
      NewPop <- mixRandomly(basePop)  
  
      ## calculate the discrepancy and counts  
      Map(function(k) {  
        Dk <- discrepancies[[k]](NewPop)  
        if (Dk >= observedVals[k])  
          counts[k] <- counts[k] + 1  
      }, 1:K)  
      counts  
    }, 1:M, init = numeric(length = K))  
  
    PVs <- total/M  
    min(PVs)  
  }  
  
  PVmin <- getPVmin(pop, discrepancies, M_inner)  
  
  total <- Reduce(function(count, m) {  
    basePop <- mixRandomly(pop)  
    if (getPVmin(basePop, discrepancies, M_inner) <= PVmin)  
      count + 1 else count  
  }, 1:M_outer, init = 0)  
  
  PVstar <- total/M_outer  
  return(PVstar)  
}
```

```

mixRandomly <- function(pop) {
  pop1 <- pop$pop1
  n_pop1 <- nrow(pop1)

  pop2 <- pop$pop2
  n_pop2 <- nrow(pop2)

  mix <- rbind(pop1, pop2)
  select4pop1 <- sample(1:(n_pop1 + n_pop2), n_pop1, replace = FALSE)

  new_pop1 <- mix[select4pop1, ]
  new_pop2 <- mix[-select4pop1, ]
  list(pop1 = new_pop1, pop2 = new_pop2)
}

```

```

bootstrap_t_interval <- function(S, a, confidence, B, D) {
  ## Inputs:
  ##   S = an n element array containing the variate values in the sample
  ##   a = a scalar-valued function that calculates the attribute a() of interest
  ##   confidence = a value in (0,1) indicating the confidence level
  ##   B = a numeric value representing the outer bootstrap count of
  ##   replicates (used to calculate the lower and upper limits)
  ##   D = a numeric value representing the inner bootstrap count of replicates
  ##   (used to estimate the standard deviation of the sample attribute for
  ##   each (outer) bootstrap sample)

  Pstar <- S
  aPstar <- a(Pstar)
  sampleSize <- length(S)
  ## get (outer) bootstrap values
  bVals <- sapply(1:B, FUN = function(b) {
    Sstar <- sample(Pstar, sampleSize, replace = TRUE)
    aSstar <- a(Sstar)
    ## get (inner) bootstrap values to estimate the SD
    Pstarstar <- Sstar
    SD_aSstar <- sd(sapply(1:D, FUN = function(d) {
      Sstarstar <- sample(Pstarstar, sampleSize, replace = TRUE)
      ## return the attribute value
      a(Sstarstar)
    })))
    z <- (aSstar - aPstar)/SD_aSstar
    ## Return the two values
    c(aSstar = aSstar, z = z)
  })
  SDhat <- sd(bVals["aSstar", ])
  zVals <- bVals["z", ]
  ## Now use these zVals to get the lower and upper c values.
  cValues <- quantile(zVals, probs = c((1 - confidence)/2, (confidence +
    1)/2), na.rm = TRUE)
  cLower <- min(cValues)
  cUpper <- max(cValues)
  interval <- c(lower = aPstar - cUpper * SDhat, middle = aPstar, upper = aPstar -
    cLower * SDhat)
  return(interval)
}

```